Muhammad Samir Assawalhy

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Education

Zagazig University Zagazig, Egypt

Bachelor of Electrical Engineering - Computer and Systems Engineering track *Sep 2019 – Jul 2024*

- Achieved 1st place among university teams in the ECPC competitions for three consecutive years

Experience

Backend Intern Jun 2026 – Sep 2026

Taager *Cairo, Egypt*

- Designed and maintained microservices communicating via Apache Pulsar
- Maintained backend systems built with Node.js and Spring Boot Kotlin, working across both SQL (PostgreSQL) and NoSQL (MongoDB) databases
- Applied Clean Architecture, DDD, and CQRS to maintain code modularity and enable parallel work in an agile, fast-ship environment
- Achieved 80% unit test coverage and wrote integration tests using Testcontainers and MockK
- Gained working knowledge of CI/CD pipelines (ArgoCD), container orchestration (Kubernetes), and monitoring and observability (Datadog)
- Leveraged feature flags to progressively roll out features and run A/B experiments

Frontend Engineer Aug 2023 – Jun 2024

SciCave *Cairo, Egypt*

- Built web apps using TypeScript, React.js, Next.js, Tailwind CSS, and other web technologies
- Tested the frontend using Cypress and CI/CD pipelines to automate quality checks and testing
- Managed deployment using Docker and cloud services like AWS EC2 machine
- Designed UI/UX using Figma and collaborated on requirements gathering

Projects

Arabi.js | *JavaScript, PEG.js* Oct 2020 – May 2021

- JavaScript Arabization based on Babel.js parser

Rasm | *JavaScript, HTML, Sass, Pugjs, PEG.js, Webpack, Gulp, Jest* Feb 2020 – Present

- Graphing calculator to plot math functions, draw geometric shapes, and animate them

Da7ee7 Bots | *Python, JavaScript, HTML, Bootstrap* Apr 2022 – May 2023

- Dashboard built with Python to automate sending messages to multiple Telegram bots

Achievements

ACPC Finalist 2022, 2023, 2024

Qualified to the African and Arab Collegiate Programming Championship

NASA Space Apps Challenge Nov 2023

My team Pyrogurarians got 3rd place and became a global nominee

IEEEXtreme 17.0 Competition Oct 2023

My team was one of the top 15 teams out of 198 Egyptian teams

Extra-Curricular Activities

Mentor and Instructor Mar 2022 – Aug 2024

ICPC Community at Zagazig University

- Mentored multiple teams in the community and managed to leverage their algorithms and problem-solving skills
- Delivered weekly sessions, covering topics like greedy algorithms and C++ STL library
- Authored problems to be used in local or regional ICPC contests and tested other problems written by colleagues

IT Committee Head Nov 2021 – Jun 2023

ZagEng Team

- Directed the committee to build 13+ Telegram bots that help students get their learning materials
- Automated repeated tasks by building a dashboard using Python and integrating it with Make.com

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Kotlin, C++, SQL, Bash
Frontend: React.js, Next.js, Tailwind CSS, Cypress, HTML, CSS
Backend: Node.js, NestJS, Spring Boot, REST APIs, WebSockets, Apache Pulsar, PostgreSQL, MongoDB
Tools: Git, Docker, Kubernetes, Linux, AWS, CI/CD (ArgoCD), Datadog, Figma
Others: Clean Architecture, DDD, CQRS, OOP, API Design, Problem Solving